

Mux, decoder, encoder

A multiplexer picks one data input under select control, two inputs need one select bit, four inputs need two

Select, one-hot, encode

- Two-to-one mux: Y equals D zero when S is zero, D one when S is one
- Four-to-one extends the select field
- A two-to-four decoder lights exactly one Y line for addr zero zero through one one
- Priority encoder
- All inputs zero means V equals zero

Browser lab

Digital Design and Verification Platform

HomeTools

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INTERACTIVE TOOL

Mux / decoder / encoder explorer

Toggle select and data lines on multiplexers (2:1 through 16:1), watch one-hot decoder outputs (2→4 through 4→16), and resolve priority encoding (4→2 / 8→3).

Starter example: 2:1 mux with S=0, D0=1, D1=0 → Y=1 (passes D0).

Challenge 0 / 28 cleared

2:1 pick D0: Set S=0 with D0=1, D1=0 so Y=1 (select D0).

Show hintCheckNextLoad challenge modeIdle

2:1 pick D02:1 pick D12:1 force Y=0Quiz: 2:1 formula4:1 select D24:1 select D34:1 Y=0

Quiz: 4:1 width2+4: Y02+4: Y32+4: Y2Quiz: one-hot3+8: Y53+8: Y73+8: Y1Quiz: 3+8 lines

Encode I0Encode I3Priority winsNo requestQuiz: priorityQuiz: mux vs decoder8:1 pick D516:1 pick D12

4+16: Y10Quiz: 4+16 lines8+3 encode I78+3 priority

Explorer

Load starter example

BLOCK

2:1 mux

S=0D0=1D1=0

Mux decoder starter

Workbook practice

- In the workbook track, write the two-to-one mux truth table by hand
- For addr one zero on a two-to-four decoder, say which Y line is high
- Encode only I three set and give Y and V
- Name one pitfall: confusing mux select width with decoder output count

Pitfalls to watch

- Do not tie multiple decoder outputs high at once, that is not one-hot
- Encoder priority rules differ by design, know high versus low index first
- And remember: the browser lab is literacy
- Real buses still need enables, tri-states, and timing on select changes

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, sketch one mux and one decoder example
- When you are ready, take the short quiz, then continue to priority and compare